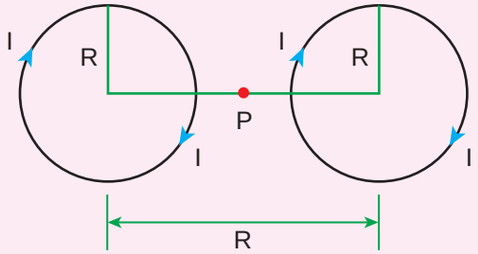
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magnetic field at a point P which is at exactly at $\frac{R}{2}$ distance between two coils is



(a) $\frac{8N\mu_{\circ}I}{\sqrt{5}R}$ (b) $\frac{8N\mu_{\circ}I}{5^{3/2}R}$

(c) $\frac{8N\mu_{\circ}I}{5R}$ (d) $\frac{4N\mu_{\circ}I}{\sqrt{5}R}$

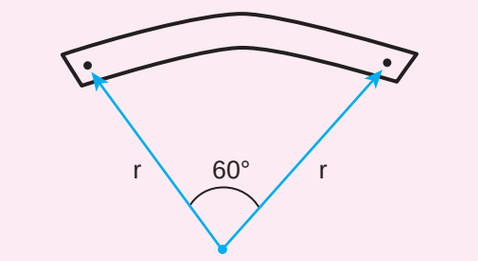
8. A wire of length l carries a current I along the Y direction and magnetic field is given by $\vec{B}=\frac{\beta}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})T$. The magnitude of Lorentz force acting on the wire is

(a) $\sqrt{\frac{2}{\sqrt{3}}}\beta Il$ (b) $\sqrt{\frac{1}{\sqrt{3}}}\beta Il$

(c) $\sqrt{2}\beta Il$ (d) $\sqrt{\frac{1}{2}}\beta Il$

9. A bar magnet of length l and magnetic moment M is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be

(NEET 2014)



(a) M (b) $\frac{3}{\pi}M$

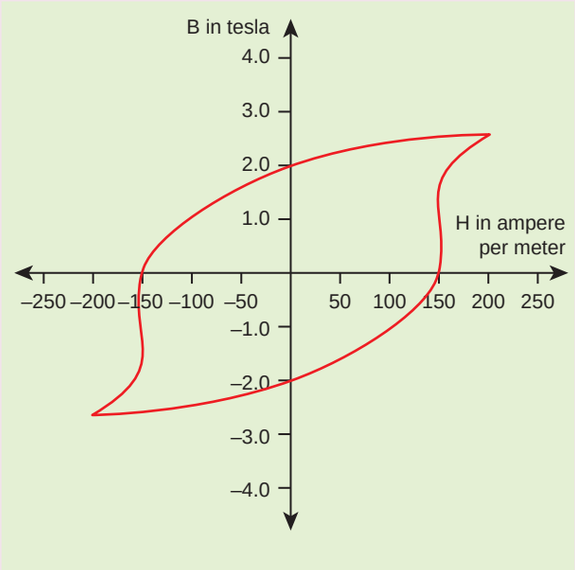
(c) $\frac{2}{\pi}M$ (d) $\frac{1}{2}M$

10. A non-conducting charged ring of charge q, mass m and radius r is rotated with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is

(a) $\frac{q}{m}$ (b) $\frac{2q}{m}$

(c) $\frac{q}{2m}$ (d) $\frac{q}{4m}$

11. The BH curve for a ferromagnetic material is shown in the figure. The material is placed inside a long solenoid which contains 1000 turns/ cm. The current that should be passed in the solenonid to demagnetize the ferromagnet completely is



(a) 1.00 m A (milli ampere) (b) 1.25 mA

(c) 1.50 mA (d) 1.75 mA

12. Two short bar magnets have magnetic moments 1.20 Am^2 and 1.00 Am^2

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